Kavimalar D 241801118 Experiment 1

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

from sklearn.preprocessing import LabelEncoder
from sklearn.preprocessing import StandardScaler
```

```
df = pd.read_csv("/content/WA_Fn-UseC_-Telco-Customer-Churn.csv")
print("First 10 Records:")
print(df.head(10))
print("\nLast 10 Records:")
print(df.tail(10))

print("\nShape of Dataset:")
print(df.shape)

print("\nColumn Names:")
print(df.columns)
```

```
First 10 Records:
   customerID  gender  SeniorCitizen Partner Dependents  tenure PhoneService  \
0  7590-VHVEG  Female              0     Yes         No       1           No
1  5575-GNVDE    Male              0      No         No      34          Yes
2  3668-QPYBK    Male              0      No         No       2          Yes
3  7795-CFOCW    Male              0      No         No      45           No
4  9237-HQITU  Female              0      No         No       2          Yes
5  9305-CDSKC  Female              0      No         No       8          Yes
6  1452-KIOVK    Male              0      No        Yes      22          Yes
7  6713-OKOMC  Female              0      No         No      10           No
8  7892-POOKP  Female              0     Yes         No      28          Yes
9  6388-TABGU    Male              0      No        Yes      62          Yes

      MultipleLines InternetService OnlineSecurity  ... DeviceProtection  \
0  No phone service             DSL             No  ...               No
1                No             DSL            Yes  ...              Yes
2                No             DSL            Yes  ...               No
3  No phone service             DSL            Yes  ...              Yes
4                No     Fiber optic             No  ...               No
5               Yes     Fiber optic             No  ...              Yes
6               Yes     Fiber optic             No  ...               No
7  No phone service             DSL            Yes  ...               No
8               Yes     Fiber optic             No  ...              Yes
9                No             DSL            Yes  ...               No

  TechSupport StreamingTV StreamingMovies        Contract PaperlessBilling  \
0          No          No              No  Month-to-month              Yes
1          No          No              No        One year               No
2          No          No              No  Month-to-month              Yes
3         Yes          No              No        One year               No
4          No          No              No  Month-to-month              Yes
5          No         Yes             Yes  Month-to-month              Yes
6          No         Yes              No  Month-to-month              Yes
7          No          No              No  Month-to-month               No
8         Yes         Yes             Yes  Month-to-month              Yes
9          No          No              No        One year               No

               PaymentMethod MonthlyCharges  TotalCharges Churn
0           Electronic check          29.85         29.85    No
1               Mailed check          56.95        1889.5    No
2               Mailed check          53.85        108.15   Yes
3  Bank transfer (automatic)          42.30       1840.75    No
4           Electronic check          70.70        151.65   Yes
5           Electronic check          99.65         820.5   Yes
6    Credit card (automatic)          89.10        1949.4    No
7               Mailed check          29.75         301.9    No
8           Electronic check         104.80       3046.05   Yes
9  Bank transfer (automatic)          56.15       3487.95    No

[10 rows x 21 columns]

Last 10 Records:
      customerID  gender  SeniorCitizen Partner Dependents  tenure  \
7033  9767-FFLEM    Male              0      No         No      38
7034  0639-TSIQW  Female              0      No         No      67
7035  8456-QDAVC    Male              0      No         No      19
7036  7750-EYXWZ  Female              0      No         No      12
7037  2569-WGERO  Female              0      No         No      72
```

```
print("Dataset Information:")
df.info()
```

```
print("\nSummary Statistics:")
print(df.describe())

print("\nNumber of Missing Values:")
print(df.isnull().sum())

print("\nNumber of Duplicate Records:")
print(df.duplicated().sum())
```

```
Dataset Information:
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 7043 entries, 0 to 7042
Data columns (total 21 columns):
 #   Column            Non-Null Count  Dtype
---  ------            --------------  -----
 0   customerID        7043 non-null   object
 1   gender            7043 non-null   object
 2   SeniorCitizen     7043 non-null   int64
 3   Partner           7043 non-null   object
 4   Dependents        7043 non-null   object
 5   tenure            7043 non-null   int64
 6   PhoneService      7043 non-null   object
 7   MultipleLines     7043 non-null   object
 8   InternetService   7043 non-null   object
 9   OnlineSecurity    7043 non-null   object
 10  OnlineBackup      7043 non-null   object
 11  DeviceProtection  7043 non-null   object
 12  TechSupport       7043 non-null   object
 13  StreamingTV       7043 non-null   object
 14  StreamingMovies   7043 non-null   object
 15  Contract          7043 non-null   object
 16  PaperlessBilling  7043 non-null   object
 17  PaymentMethod     7043 non-null   object
 18  MonthlyCharges    7043 non-null   float64
 19  TotalCharges      7043 non-null   object
 20  Churn             7043 non-null   object
dtypes: float64(1), int64(2), object(18)
memory usage: 1.1+ MB

Summary Statistics:
       SeniorCitizen       tenure  MonthlyCharges
count    7043.000000  7043.000000     7043.000000
mean        0.162147    32.371149       64.761692
std         0.368612    24.559481       30.090047
min         0.000000     0.000000       18.250000
25%         0.000000     9.000000       35.500000
50%         0.000000    29.000000       70.350000
75%         0.000000    55.000000       89.850000
max         1.000000    72.000000      118.750000

Number of Missing Values:
customerID          0
gender              0
SeniorCitizen       0
Partner             0
Dependents          0
tenure              0
PhoneService        0
MultipleLines       0
InternetService     0
OnlineSecurity      0
OnlineBackup        0
DeviceProtection    0
TechSupport         0
StreamingTV         0
StreamingMovies     0
Contract            0
```

```
df["TotalCharges"] = pd.to_numeric(df["TotalCharges"], errors="coerce")

df.drop_duplicates(inplace=True)

numerical_cols = df.select_dtypes(include=np.number).columns
categorical_cols = df.select_dtypes(include="object").columns

for col in numerical_cols:
    df[col] = df[col].fillna(df[col].median())

for col in categorical_cols:
    df[col] = df[col].fillna(df[col].mode()[0])

print("Missing Values After Cleaning:")
print(df.isnull().sum())
```

```
Missing Values After Cleaning:
customerID          0
```

```
gender              0
SeniorCitizen       0
Partner             0
Dependents          0
tenure              0
PhoneService        0
MultipleLines       0
InternetService     0
OnlineSecurity      0
OnlineBackup        0
DeviceProtection    0
TechSupport         0
StreamingTV         0
StreamingMovies     0
Contract            0
PaperlessBilling    0
PaymentMethod       0
MonthlyCharges      0
TotalCharges        0
Churn               0
dtype: int64
```

```
# Encode categorical columns

encode_cols = [
    "gender",
    "Partner",
    "Dependents",
    "Contract",
    "PaymentMethod",
    "InternetService",
    "Churn"
]

le = LabelEncoder()

for col in encode_cols:
    df[col] = le.fit_transform(df[col])

print("Dataset After Encoding:")
print(df.head())
```

```
Dataset After Encoding:
   customerID  gender  SeniorCitizen  Partner  Dependents  tenure  \
0  7590-VHVEG       0              0        1           0       1
1  5575-GNVDE       1              0        0           0      34
2  3668-QPYBK       1              0        0           0       2
3  7795-CFOCW       1              0        0           0      45
4  9237-HQITU       0              0        0           0       2

  PhoneService     MultipleLines  InternetService OnlineSecurity  ...  \
0           No  No phone service                0             No  ...
1          Yes                No                0            Yes  ...
2          Yes                No                0            Yes  ...
3           No  No phone service                0            Yes  ...
4          Yes                No                1             No  ...

  DeviceProtection TechSupport StreamingTV StreamingMovies Contract  \
0               No          No          No              No        0
1              Yes          No          No              No        1
2               No          No          No              No        0
3              Yes         Yes          No              No        1
4               No          No          No              No        0

   PaperlessBilling PaymentMethod  MonthlyCharges  TotalCharges  Churn
0               Yes             2           29.85         29.85      0
1                No             3           56.95       1889.50      0
2               Yes             3           53.85        108.15      1
3                No             0           42.30       1840.75      0
4               Yes             2           70.70        151.65      1

[5 rows x 21 columns]
```

```
# Standardise numerical features

scale_cols = [
    "tenure",
    "MonthlyCharges",
    "TotalCharges"
]

scaler = StandardScaler()

df[scale_cols] = scaler.fit_transform(df[scale_cols])
```

```
print("Standardised Numerical Features:")
print(df[scale_cols].head())
```

```
Standardised Numerical Features:
     tenure  MonthlyCharges  TotalCharges
0 -1.277445       -1.160323     -0.994242
1  0.066327       -0.259629     -0.173244
2 -1.236724       -0.362660     -0.959674
3  0.514251       -0.746535     -0.194766
4 -1.236724        0.197365     -0.940470
```

```
# Distribution of MonthlyCharges

plt.figure(figsize=(8, 5))
sns.histplot(df["MonthlyCharges"], kde=True)

plt.title("Distribution of Monthly Charges")
plt.xlabel("Monthly Charges")
plt.ylabel("Frequency")

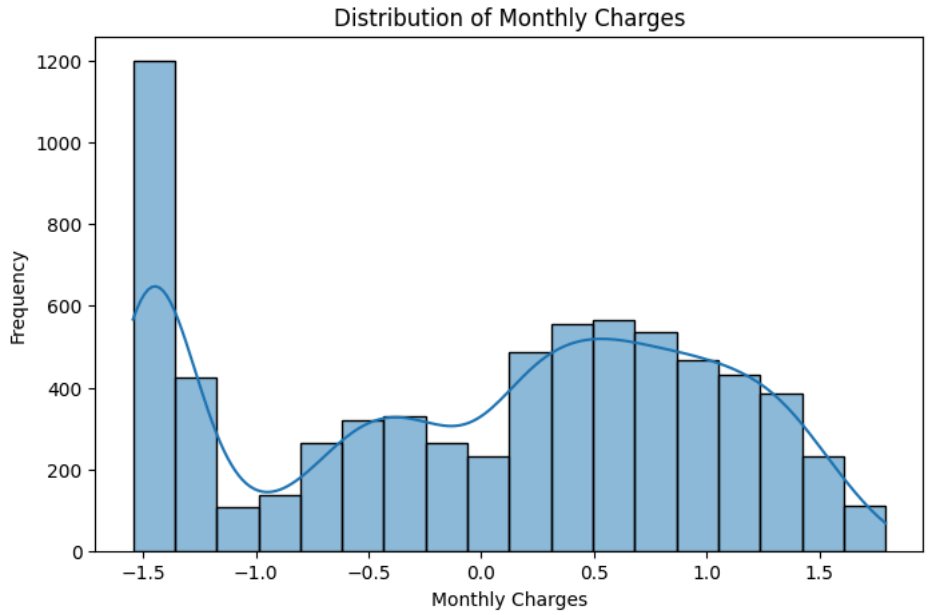
plt.show()

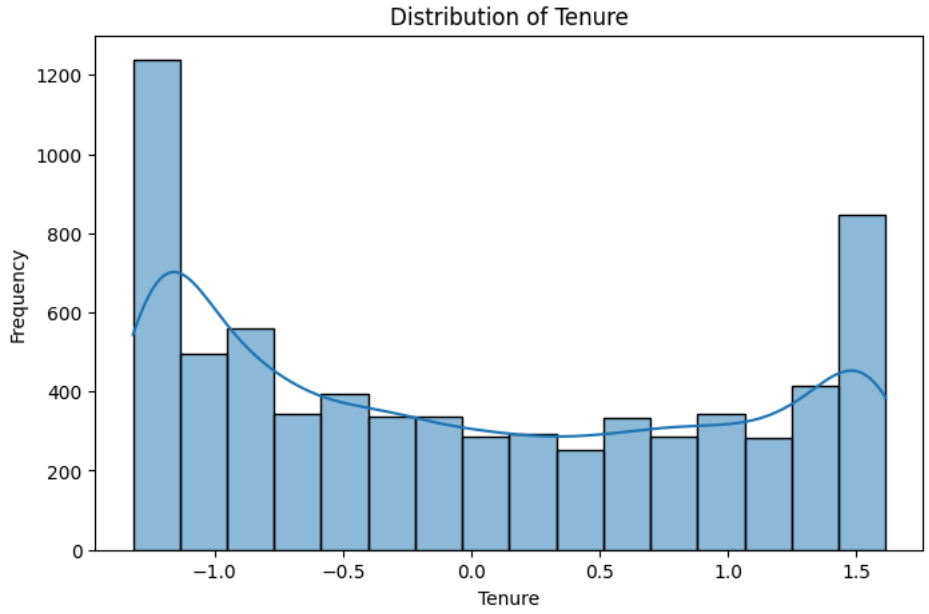

# Distribution of Tenure

plt.figure(figsize=(8, 5))
sns.histplot(df["tenure"], kde=True)

plt.title("Distribution of Tenure")
plt.xlabel("Tenure")
plt.ylabel("Frequency")

plt.show()
```





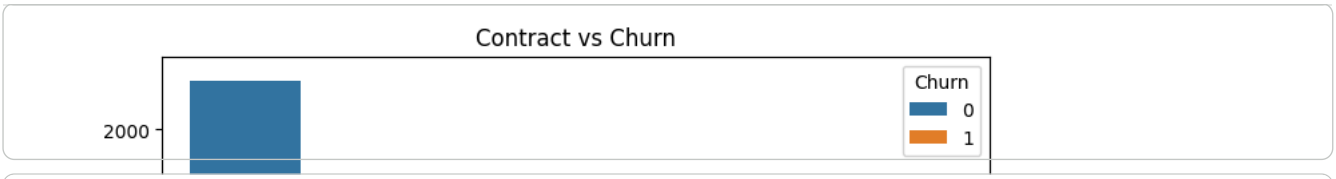

```
# Count Plot - Contract vs Churn

plt.figure(figsize=(8, 5))

sns.countplot(data=df, x="Contract", hue="Churn")

plt.title("Contract vs Churn")
plt.xlabel("Contract")
plt.ylabel("Count")

plt.show()
```



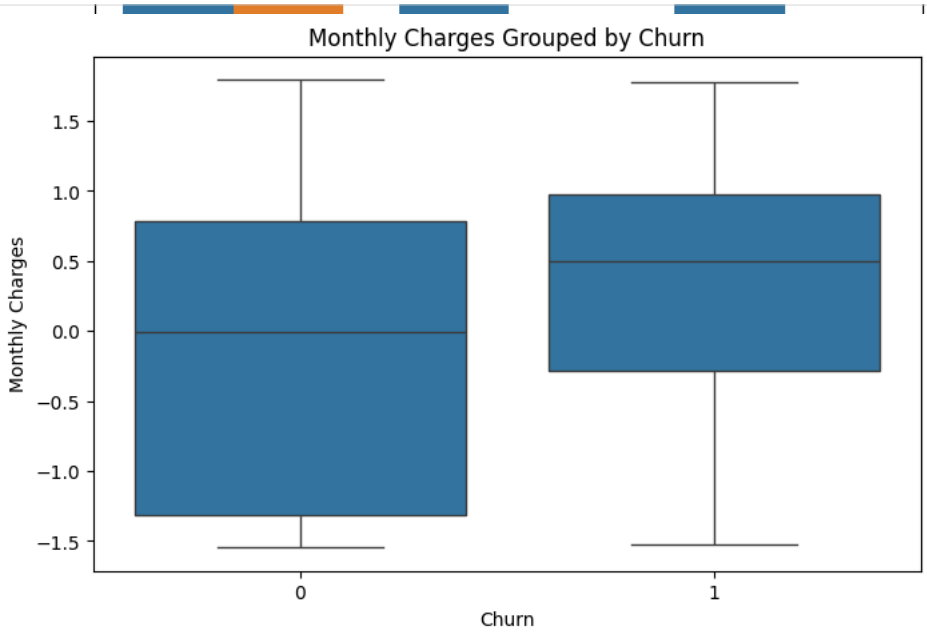

```
# Box Plot of MonthlyCharges grouped by Churn

plt.figure(figsize=(8, 5))

sns.boxplot(data=df, x="Churn", y="MonthlyCharges")

plt.title("Monthly Charges Grouped by Churn")
plt.xlabel("Churn")
plt.ylabel("Monthly Charges")

plt.show()
```




```
# Correlation Heatmap

plt.figure(figsize=(12, 8))

correlation = df.select_dtypes(include=np.number).corr()

sns.heatmap(
    correlation,
    annot=True,
    cmap="coolwarm",
    fmt=".2f"
)

plt.title("Correlation Heatmap of Numerical Features")

plt.show()
```

